# API Documentation: AI-Powered Real Estate Property Valuation Assistant

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**Domain**: Real Estate & PropTech Valuation Intelligence

**Capstone**: Prodapt AI/ML Engineering Capstone

Base URL: `http://localhost:8000`

API Prefix: `/api/v1`

Interactive OpenAPI UI: `http://localhost:8000/docs`

Total Endpoints: **16 REST Endpoints**

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## 1. Root System Status

- **Endpoint**: `/`
- **HTTP Method**: `GET`
- **Description**: Returns microservice identity, operational status, and interactive documentation link.
- **Request Format**: None
- **Sample Request**:

**Code:**
```
curl -X GET http://localhost:8000/
```

- **Sample Response**:

**Code:**
```
{
"status": "online",
"service": "AI-Powered Real Estate Property Valuation Assistant",
"docs": "/docs"
}
```

---

## 2. System Health Check

- **Endpoint**: `/health`
- **HTTP Method**: `GET`
- **Description**: Returns microservice operational health status and version identifier.
- **Request Format**: None
- **Sample Request**:

**Code:**
```
curl -X GET http://localhost:8000/health
```

- **Sample Response**:

**Code:**
```
{
"status": "healthy",
"service": "AI-Powered Real Estate Property Valuation Assistant",
"version": "1.0.0"
}
```

---

## 3. Input Validation Guardrails

- **Endpoint**: `/api/v1/guardrails/validate`
- **HTTP Method**: `POST`
- **Description**: Validates structured property features and natural-language user queries against physical boundaries (14 fields: living area 100–15,000 sq ft, quality 1–10, etc.) and detects prompt injection attacks.
- **Request Format**:

```
Code:
{
"features": {
"neighborhood": "North Ames",
"gr_liv_area": 1800,
"bedrooms": 3,
"bathrooms": 2,
"year_built": 1995,
"overall_qual": 7,
"overall_cond": 6,
"total_rooms": 7,
"garage_yr_blt": 1995
},
"query": "3 bedroom house in North Ames"
}
```

- **Sample Response**:

```
Code:
{
"is_valid": true,
"sanitized_input": {
"neighborhood": "North Ames",
"gr_liv_area": 1800.0,
"bedrooms": 3,
"bathrooms": 2.0,
"year_built": 1995,
"overall_qual": 7,
"overall_cond": 6,
"total_rooms": 7,
"garage_yr_blt": 1995
},
"errors": [],
"warnings": []
}
```

## 4. Hybrid Search & Comparable Prioritization

- **Endpoint**: `/api/v1/search`
- **HTTP Method**: `POST`
- **Description**: Executes true hybrid retrieval combining dense semantic embeddings (`all-MiniLM-L6-v2`, 384 dimensions) and lexical TF-IDF keyword search over the full candidate set (alpha = 0.55). Supports metadata filtering, real-sales isolation (`include_synthetic=False` by default), and dynamic neighborhood widening.
- **Empty Filter Behavior**: When metadata filters match 0 properties, returns an empty list (`[]`) with `count: 0` instead of falling back to all properties.
- **Request Format**:

**Code:**
```
{
"query": "3 bedroom single family home in North Ames built after 1990",
"top_k": 3,
"filters": {
"neighborhood": "North Ames",
"min_bedrooms": 3,
"min_year_built": 1990
},
"target_property": {
"neighborhood": "North Ames",
"gr_liv_area": 1800,
"bedrooms": 3,
"bathrooms": 2,
"year_built": 1995,
"overall_qual": 7,
"total_rooms": 7,
"garage_yr_blt": 1995
}
}
```

- **Sample Response**:

**Code:**
```
{
"query": "3 bedroom single family home in North Ames built after 1990",
"count": 3,
"filters_applied": {
"neighborhood": "North Ames",
"min_bedrooms": 3,
"min_year_built": 1990
},
"comparables": [
{
"property_id": "PROP-2619",
"neighborhood": "North Ames",
"sale_price": 172000.0,
"living_area": 1656,
"bedrooms": 3,
"bathrooms": 2.0,
"year_built": 1996,
"overall_quality": 6,
"hybrid_score": 0.8124,
"comparability_score": 0.8841,
"final_score": 0.8483
}
]
}
```

## 5. Property Classification Agent

- **Endpoint**: `/api/v1/classify`
- **HTTP Method**: `POST`
- **Description**: Segments property profiles into Market Tiers, Architectural Typologies, Condition Preservation Grades, and Target Demographics.
- **Request Format**:

**Code:**
```
{
"property_features": {
"neighborhood": "North Ames",
"gr_liv_area": 1800,
"bedrooms": 3,
"bathrooms": 2,
"year_built": 1995,
"overall_qual": 7,
"overall_cond": 6
}
}
```

- **Sample Response**:

**Code:**
```
{
"market_tier": "Upper-Middle Market",
"typology": "Suburban Single-Family",
"preservation_grade": "Grade B (Standard Modern)",
"target_buyer_demographic": "Move-up families and professional households",
"liquidity_assessment": "High liquidity (core suburban segment with active demand)"
}
```

## 6. Basic Valuation Recommendation Agent

- **Endpoint**: `/api/v1/valuation/basic`
- **HTTP Method**: `POST`
- **Description**: Calculates comp-adjusted median, mean, and size-weighted PPSF fair market value using retrieved historical comps with quality delta adjustments.
- **Request Format**:

**Code:**
```
{
"property_features": {
"neighborhood": "North Ames",
"gr_liv_area": 1600,
"bedrooms": 3,
"bathrooms": 2,
"year_built": 1970,
"overall_qual": 6,
"overall_cond": 5
},
"top_k_comps": 5
}
```

- **Sample Response**:

**Code:**
```
{
"estimated_fair_market_value": 171200.0,
"valuation_range_low": 136960.0,
"valuation_range_high": 205440.0,
"median_comparable_price": 171200.0,
"mean_comparable_price": 170733.33,
"implied_price_per_sqft": 107.0,
"comparable_count_used": 3,
"confidence_level": "High (Strong comparable alignment)",
"valuation_explanation": "Based on 3 prioritized comparable sales in North Ames, the median comparable value
is $171,200.00."
}
```

## 7. Supervised ML Price Prediction

- **Endpoint**: `/api/v1/valuation/predict-price`
- **HTTP Method**: `POST`
- **Description**: Computes statistical price prediction using a trained `GradientBoostingRegressor` (R^2 = 0.9198, RMSE = \25,361.84$) with 90% empirical prediction intervals.
- **Request Format**:

```
Code:
{
"property_features": {
"neighborhood": "North Ames",
"gr_liv_area": 1600,
"bedrooms": 3,
"bathrooms": 2,
"year_built": 1970,
"overall_qual": 6,
"overall_cond": 5,
"garage_cars": 2,
"total_bsmt_sf": 1000
}
}
```

- **Sample Response**:

```
Code:
{
"predicted_price": 172374.0,
"lower_bound_90pct": 130689.86,
"upper_bound_90pct": 214058.14,
"price_per_sqft": 107.73,
"model_r2_score": 0.9198,
"model_used": "GradientBoostingRegressor",
"input_features": {
"OverallQual": 6.0,
"OverallCond": 5.0,
"GrLivArea": 1600.0,
"TotalBsmtSF": 1000.0,
"GarageCars": 2.0,
"YearBuilt": 1970.0
}
}
```

---

## 8. Property Value Risk Assessment

- **Endpoint**: `/api/v1/valuation/predict-risk`
- **HTTP Method**: `POST`
- **Description**: Assesses risk (0–100 index) across Physical Age Depreciation (30%), Quality-Condition Discrepancy (25%), Neighborhood Price Volatility (25%), and Layout Obsolescence (20%).
- **Request Format**:

```
Code:
{
"property_features": {
"neighborhood": "North Ames",
"gr_liv_area": 1600,
"bedrooms": 3,
"bathrooms": 2,
"year_built": 1970,
"overall_qual": 6,
"overall_cond": 5
}
}
```

- **Sample Response**:

```
Code:
{
"composite_risk_score": 41.3,
"risk_tier": "Moderate Risk",
"breakdown": {
"physical_depreciation_risk": 56.0,
"condition_gap_risk": 20.0,
"market_volatility_risk": 35.0,
"liquidity_risk": 40.0
},
"key_risk_drivers": [
"Structure is 56 years old; mechanical systems may require capital reserves."
]
}
```

## 9. Feature Impact & What-If Renovation Analysis

- **Endpoint**: `/api/v1/valuation/impact-analysis`
- **HTTP Method**: POST
- **Description**: Computes neighborhood price premiums relative to the city median ($160,000) and calculates real-time valuation changes from planned renovations.
- **Request Format**:

```
Code:
{
"neighborhood": "North Ames",
"base_features": {
"neighborhood": "North Ames",
"gr_liv_area": 1600,
"bedrooms": 3,
"bathrooms": 2,
"year_built": 1970,
"overall_qual": 6,
"overall_cond": 5
},
"renovation_adjustments": {
"added_sqft": 300,
"quality_increase": 1
}
}
```

- **Sample Response**:

**Code:**
```
{
"neighborhood_analysis": {
"neighborhood": "North Ames",
"median_price": 140000.0,
"city_median_price": 160000.0,
"premium_or_discount_pct": -12.5,
"tier": "Mid-Market Value"
},
"what_if_simulation": {
"baseline_valuation": 172374.0,
"renovated_valuation": 205840.0,
"value_lift_dollars": 33466.0,
"value_lift_percentage": 19.42
}
}
```

## 10. LLM-as-Judge Valuation Audit

- **Endpoint**: `/api/v1/valuation/judge-validate`
- **HTTP Method**: `POST`
- **Description**: Audits valuation consistency against historical comps and ML models, returning structured verdicts (`APPROVED`, `FLAGGED_OVERVALUED`, `FLAGGED_UNDERVALUED`, `NEEDS_REVIEW`). Rejects invalid inputs with HTTP 400.
- **Request Format**:

**Code:**
```
{
"reconciled_fmv": 171631.12,
"comp_median": 171200.0,
"ml_prediction": 172374.0,
"confidence_interval": {
"low": 131616.53,
"high": 211645.71
},
"property_features": {
"neighborhood": "North Ames",
"gr_liv_area": 1600,
"overall_qual": 6
}
}
```

- **Sample Response**:

**Code:**
```
{
"verdict": "APPROVED",
"audit_passed": true,
"confidence_score": 0.95,
"discrepancy_percentage": 0.43,
"audit_rationale": "The reconciled valuation of $171,631.12 aligns closely with both the comp median
($171,200.00) and ML point estimate ($172,374.00), well within the 90% confidence interval."
}
```

## 11. Multi-Agent Valuation Orchestrator

- **Endpoint**: `/api/v1/valuation/collaborate`
- **HTTP Method**: `POST`
- **Description**: Orchestrates the complete multi-agent workflow: guardrails, classification, hybrid retrieval, dynamic `comp_fit` reconciliation, risk scoring, feature analysis, and LLM-as-Judge auditing.

- **Request Format**:

```
Code:
{
"property_features": {
"neighborhood": "North Ames",
"gr_liv_area": 1600,
"bedrooms": 3,
"bathrooms": 2,
"year_built": 1970,
"overall_qual": 6,
"overall_cond": 5,
"garage_cars": 2,
"total_rooms": 7,
"garage_yr_blt": 1970
},
"query": "Standard 3 bedroom home in North Ames built in 1970",
"top_k_comps": 5
}
```

- **Sample Response**:

```
Code:
{
"reconciled_fair_market_value": 171631.12,
"confidence_interval": {
"low": 131616.53,
"high": 211645.71
},
"comp_fit": 1.0,
"comp_weight": 0.5,
"ml_weight": 0.5,
"widened_search_used": false,
"classification": {
"market_tier": "Mid-Market Standard",
"typology": "Suburban Single-Family"
},
"basic_valuation": {
"median_comparable_price": 171200.0,
"comparables_summary": [
{"property_id": "PROP-2619", "sale_price": 172000.0},
{"property_id": "PROP-1613", "sale_price": 171200.0},
{"property_id": "PROP-1928", "sale_price": 169000.0}
]
},
"ml_prediction": {
"predicted_price": 172374.0,
"model_r2_score": 0.9198
},
"risk_assessment": {
"composite_risk_score": 41.3,
"risk_tier": "Moderate Risk"
},
"judge_audit": {
"verdict": "APPROVED",
"audit_passed": true
}
}
```

## 12. Continuous Improvement: Feedback Submission

- **Endpoint**: `/api/v1/feedback/submit`
- **HTTP Method**: `POST`

- **Description**: Captures appraiser evaluations, actual verified sales prices, user ratings (1–5 stars), and optional applied calibration factor to update model calibration metrics.
- **Request Format**:

**Code:**
```
{
"property_id": "PROP-2619",
"recommended_value": 171631.12,
"actual_sale_price": 172000.0,
"user_rating": 5,
"applied_calibration_factor": 1.0,
"appraiser_comments": "Valuation aligned closely with executed purchase contract.",
"professional_role": "Certified General Appraiser"
}
```

- **Sample Response**:

**Code:**
```
{
"feedback_id": "FB-20261011-849201",
"status": "recorded",
"error_dollars": 368.88,
"error_pct": 0.21,
"calibration_applied": 1.0
}
```

---

## 13. Continuous Improvement: Summary & Calibration Bias

- **Endpoint**: `/api/v1/feedback/summary`
- **HTTP Method**: `GET`
- **Description**: Returns aggregate feedback learning metrics, MAE, MAPE, uncalibrated systematic bias, and active dynamic calibration factor.
- **Request Format**: None
- **Sample Response**:

**Code:**
```
{
"total_feedback_count": 5,
"average_rating": 4.8,
"mean_absolute_error_dollars": 3412.5,
"mean_absolute_percentage_error": 1.95,
"systematic_bias_pct": 0.32,
"calibration_factor": 1.0032,
"recent_records": [...]
}
```

---

## 14. DeepEval Quality Benchmark Execution

- **Endpoint**: `/api/v1/evaluation/run`
- **HTTP Method**: `GET`
- **Description**: Executes the DeepEval recommendation quality benchmark test suite across 5 curated test scenarios (`EVAL-01` through `EVAL-05`). Operates honestly in dual modes. Automatically persists the run to `data/eval_history.json` (retaining the 3 most recent runs) and returns a `run_at` ISO timestamp in the response.
- **Request Format**: None

**Live LLM Mode Output (Measured from live execution)**:

**Code:**
```
{
"evaluation_framework": "DeepEval (v4.2.8)",
"evaluation_mode": "live_llm_judge",
"execution_time_seconds": 61.98,
"total_scenarios_evaluated": 5,
"passed_scenarios": 5,
"pass_rate_percentage": 100.0,
"mean_scores": {
"mean_faithfulness": 0.900,
"mean_answer_relevancy": 0.880,
"mean_contextual_precision": 0.850,
"mean_contextual_recall": 0.850,
"mean_valuation_accuracy": 1.000
},
"benchmark_status": "PASSED",
"run_at": "2026-10-12T12:00:00"
}
```

**Offline Mode Output (Without `OPENAI_API_KEY`)**:

**Code:**
```
{
"evaluation_framework": "DeepEval (v4.2.8)",
"evaluation_mode": "offline_no_llm",
"execution_time_seconds": 4.2,
"total_scenarios_evaluated": 5,
"passed_scenarios": null,
"pass_rate_percentage": null,
"mean_scores": {
"mean_faithfulness": null,
"mean_answer_relevancy": null,
"mean_contextual_precision": null,
"mean_contextual_recall": null,
"mean_valuation_accuracy": 1.000
},
"benchmark_status": "NOT_EVALUATED",
"run_at": "2026-10-12T12:00:00"
}
```

---

## 15. DeepEval Benchmark History

- **Endpoint**: `/api/v1/evaluation/history`
- **HTTP Method**: `GET`
- **Description**: Returns the last 3 saved DeepEval benchmark runs, ordered newest first. Returns an empty array `[]` if no runs have been executed yet or if history is missing.
- **Request Format**: None
- **Sample Request**:

**Code:**
```
curl -X GET http://localhost:8000/api/v1/evaluation/history
```

- **Sample Response**:

**Code:**

```
[
{
"evaluation_framework": "DeepEval (v4.2.8)",
"evaluation_mode": "live_llm_judge",
"execution_time_seconds": 61.98,
"total_scenarios_evaluated": 5,
"passed_scenarios": 5,
"pass_rate_percentage": 100.0,
"mean_scores": {
"mean_faithfulness": 0.9,
"mean_answer_relevancy": 0.88,
"mean_contextual_precision": 0.85,
"mean_contextual_recall": 0.85,
"mean_valuation_accuracy": 1.0
},
"benchmark_status": "PASSED",
"scenario_results": [...],
"run_at": "2026-10-12T12:00:00"
}
]
```


## 16. Dataset Information & Summary

- **Endpoint**: `/api/v1/dataset/info`
- **HTTP Method**: `GET`
- **Description**: Returns dataset documentation, record counts (including synthetic augmentation), key fields, and statistical summary metrics.
- **Request Format**: None
- **Sample Request**:

**Code:**
```
curl -X GET http://localhost:8000/api/v1/dataset/info
```

- **Sample Response**:

**Code:**

```
{
"dataset_name": "Ames Housing Dataset",
"source": "OpenML Data ID 43926 / Dean De Cock (2011)",
"total_records": 10000,
"key_fields": [
"SalePrice", "OverallQual", "OverallCond", "GrLivArea",
"TotalBsmtSF", "GarageCars", "GarageArea", "YearBuilt",
"YearRemodAdd", "FullBath", "BedroomAbvGr", "TotRmsAbvGrd",
"Neighborhood", "LotArea", "GarageYrBlt"
],
"neighborhoods_count": 28,
"price_statistics": {
"min": 12789.0,
"median": 160000.0,
"mean": 180406.95,
"max": 755000.0
}
}
```